

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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Embedded firmware engineer and computer engineering graduate with four internships spanning electrical systems and manufacturing. Developed C/C++ firmware, production tools, manufacturing databases, and plant-wide tooling records.

SKILLS

Programming & Data: C, C++, SQL, VBA, Microsoft Access, Excel, Python, MATLAB, Bash

Embedded & Controls: STM32G474, dsPIC33CK, PIC32, TI C2000/TMS320, MSP430FR2355; bare-metal firmware, FreeRTOS, Embedded Linux, ADC sensing, PWM motor control, timers

Interfaces: CAN 2.0, CAN FD, J1939, I2C, SPI, UART/RS-232, GPIO

Hardware & Manufacturing: Schematic interpretation; PCB review in Altium, KiCad, and P-CAD; JTAG, oscilloscopes, logic/protocol analyzers; production programming, test automation, tooling data

EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

May 2026–Present

- ▷ Wrote and debugged C firmware for dsPIC33CK-based power hardware, including CAN, ADC, PWM, I2C, SPI, UART, and timers; verified electrical signals and bus traffic with JTAG, oscilloscopes, and logic analyzers.
- ▷ Used schematics and PCB layouts from Altium, KiCad, and P-CAD to trace system failures across wiring, power, firmware, CAN traffic, device settings, Linux services, and application software.
- ▷ Integrated CAN-connected power devices with a TI AM62 Embedded Linux controller, including Yocto/Arago images, Linux kernels, device trees, boot troubleshooting, and an external-watchdog service.

AEGIS Power Systems | Embedded Systems Intern | Computer Science Intern

Summers 2024 & 2025

Selected work across both internships

- ▷ Wrote bare-metal C firmware for a TMS320-controlled bidirectional power converter with 12 PWM A/B pairs (24 outputs); configured frequency, duty cycle, and phase for each A output and set its paired B output to follow or invert it.
- ▷ Worked on a two-processor power system in which a FreeRTOS PIC32 handled I2C, SPI, CAN/J1939, UART, and firmware updates while a bare-metal TMS320 ran the real-time converter controls.
- ▷ Built a firmware update and recovery workflow that transferred checked image chunks over TCP, stored new and golden images in PIC32 SPI flash, and programmed and verified the TMS320 over UART.

Production and engineering tools

- ▷ Simplified TI's C++ serial flasher for the C2000 TMS320F28379D by removing unused device, CPU2, and dual-boot options, allowing production staff to program precompiled firmware without choosing engineering-only settings.
- ▷ Wrote procedures for firmware programming, test setup, device configuration, and system commissioning; built device-discovery, programming, and diagnostic tools for engineers and production staff.

TEAM Industries | Engineering Intern

Summers 2021 & 2022

- ▷ Built Excel/VBA tools that parsed inspection data, checked dimensional and spline tolerances, and flagged out-of-spec parts for engineering and manufacturing teams.
- ▷ Created and maintained a 17,000+ entry tooling where-used list for the Inrite plant, linking each tool to part numbers, operation numbers, storage locations, and vendors with custom lookup formulas and data-validation rules.

PROJECTS

Manufacturing Test Automation | SQL, VBA, Access, Excel

TEAM Industries

- ▷ Built tools for an Acceptance Test Procedure database that parsed and checked test results, retrieved earlier records, and entered new results automatically.

EV Active Sensor Adapter | Battery Monitoring & CAN

NC State University

- ▷ Wrote STM32G474 firmware for a custom EV sensor board that read ten ADC thermistors, exchanged CRC-checked cell and fault data with a BQ79616 over UART, and transmitted measurements over CAN.

C++ Microarchitecture Simulators

NC State University

- ▷ Built C++ cache, branch-predictor, and out-of-order pipeline simulators; measured cache misses, branch accuracy, memory traffic, and instructions per cycle.

TEACHING EXPERIENCE

ECE 306 Embedded Systems | Teaching Assistant & Course Mentor

Spring 2025–Spring 2026

- ▷ Taught supplemental lessons and helped students debug MSP430FR2355 firmware and custom PCBs across three semesters, including ADC sensors, PWM motor control, timers, wiring, soldering, and toolchain problems.

IEEE LEADERSHIP

IEEE at NC State | OPS2 Lead & Parts Distribution Coordinator

Fall 2025–Spring 2026

- ▷ Led two semesters of OPS2, IEEE's ten-week applied electronics course that paired classroom theory with hands-on soldering, PCB and BOM development, embedded programming, PlatformIO, and GitHub.
- ▷ Built and managed an ECE 306 parts program that cut student costs and raised \$10,000 for IEEE in two weeks.